

In this experiment, we analysed 12 acute myeloid leukaemia stem cells that were either treated with a BET inhibitor (BETi), MDM2 inhibitor (MDM2i), and a combination of both (Combo), alongside DMSO (untreated) as a control. There were three samples in each group. The samples showed an even distribution of gene expression, indicating good sample quality (Figure. 1A). There was an almost equally distribution of genes along the y axis of BETi vs DMSO MA plot having both upregulated and downregulated genes (Figure.1B). In MDM2i vs DMSO, few significant genes were observed with more upregulated and few down regulated genes. Suggesting the induction of minimal transcriptional changes by MDM2i and specific genes being targeted (Figure.1C). In Combo vs DMSO, there was an observable increased down regulation and upregulation of expressed genes, suggesting a robust transcriptome change in Combo treated cells (Figure.1D). These MA's assured the quality of the samples. The scree plot showed that approximately 70% of the total sample variance was contributed by the first two principal components (Figure.1E). PC1 explained 55.82% of the variance, while PC2 accounted for 14.74%. In the PCA plot, the samples were distributed distinctly across these two components, showing a transcriptomic shift (Figure. 1F). A Spearman correlation coefficient showed the correlation of expression of genes within the sample groups. Genes from the cells treated with an MDM2 inhibitor positively correlated with untreated samples. While the untreated samples mildly negatively correlated with the BET inhibitor-treated samples and highly negatively correlated with samples that were treated with a combination of both (Figure. 1G). Although the untreated samples correlated with the MDM2i-treated group, the PCA plot still revealed a transcriptomic separation between them (Supplementary Figure. 1A). A similar separation was observed between the BETi-treated samples and the combination-treated samples (Supplementary Figure. 1B).

Differential gene expression analysis among BETi vs DMSO, MDM2i vs DMSO and Combo vs DMSO.

In the BETi vs. control analysis, 738 genes were downregulated, and 484 were upregulated (Figure. 2A). Of the 1,222 differentially expressed genes, approximately 60% were lowly expresses in BET inhibitor–treated cells compared with the untreated, while the remaining proportion showed higher expression (Figure. 2B). This suggests that the BET inhibitor downregulates more genes than it upregulates. Among the top 10 highly expressed genes, genes responsible for the proliferation and survival of AML stem cells, like *CTSG*, *LYZ*, and *SRGN*, were notably downregulated in BET inhibitor-treated cells (Figure. 2C) (1,2). In the same treatment group, there was an up-regulation of long non-coding RNA genes like *NEAT1* that suppresses AML stem cells by affecting proliferation, activation and differentiation of the cells (3,4)(Figure. 2C). Therefore, this suggests that the mechanism of action of BETi involves inhibition of AML cell proliferation and differentiation.

Only 21 genes were differentially expressed in AML cells treated with the MDM2 inhibitor, with 18 upregulated and 3 downregulated (Figure. 2D). Overall, gene expression pattern was higher in the treated samples than in the controls (Figure. 2E), including notable increases in *MDM2* and *CDKN1A* (Figure. 2F). The elevated expression of *MDM2* appears

paradoxical; however, it is well established that MDM2 inhibitors lead to increased MDM2 transcription through p53 activation, as inhibition of MDM2 stabilises and activates p53, thereby enhancing MDM2 expression as a feedback response(5). *CDKN1A* is known for cellular senescence (6) with a dual function of causing apoptosis (7).

Cells treated with the combination of BET and MDM2 inhibitors showed comparable numbers of upregulated and downregulated genes, with 848 genes downregulated and 607 upregulated (Figure. 2G). The overall gene-expression pattern demonstrated both highly expressed genes and lowly expressed genes in the combination-treated AML stem cells (Figure. 2H). Notably, genes that were downregulated in the BET inhibitor group, such as *CTSG*, also remained downregulated in the combination treatment. Similarly, genes that were upregulated in MDM2-inhibitor-treated cells, such as *CDKN1A*, were likewise upregulated under the combination therapy (8). This is consistent with existing literature showing that combination treatment induces both upregulation and downregulation of gene sets simultaneously, reflecting the combined molecular effects of BET and MDM2 inhibition.

Affected Biological Pathways induced by BET Inhibitor, MDM2 Inhibitor, and their Combination.

To identify the biological pathways affected by the treatments, an over-representation analysis (ORA) of enriched biological processes was performed for all treatment comparison groups (Figure. 3).

In BETi-treated cells, the significantly affected pathways were primarily associated with DNA replication and cell-cycle regulation, indicating impaired cell proliferation (Supplementary Figure. 1D). Because the three upregulated genes did not alter any biological pathway, all the identified pathways were driven exclusively by the downregulated gene set (Figure. 3A). Djamai et al. also demonstrated that BET inhibitors reduce cell proliferation and induce apoptosis independently of p53 (9). This is consistent with our pathway enrichment results, where no p53-related pathways were directly enriched in the BETi group. Instead, BET inhibition affected pathways such as ribosome biogenesis (Figure 3A, Supplementary Figure. 1E), and defects in this process are known to activate p53 and promote apoptosis (10). Additionally, key cell-cycle regulatory genes within these pathways, including *FEN1*, *CDC45*, *MCM7* and *CCNA2*, were downregulated (Figure. 3B) (11). Together, these observations indicate that BETi suppresses proliferation and drives reversible cell arrest in the G0/G1 phase(12–14).

In the MDM2i treatment group, only the upregulated genes altered the biological pathways (Supplementary Figure. F and Figure 3). Intrinsic apoptosis signalling pathway mediated by p53 class mediator was the most enriched pathway(Figure. 3C), and the other ontologies were also related to apoptosis, showing that the MDM2i is linked directly to p53 cell death. All the genes in the top ontology were upregulated including *PHLDA3*, which regulates p53 activity and promotes apoptosis (15,16) (Figure. 3D)

In the ORA of genes in the combination treatment, chromosomal segregation was the most enriched pathway, having 104 overlapping genes, where about 90% of them were downregulated (Supplementary Figure. 1G, H). The affected pathways of downregulated genes related to cell division which resembled the affected pathways in the BETi group. However, cell division on chromosomal level was also observed (Figure. 3E). The ORA of upregulated genes in the combination group revealed direct enrichment of p53-related pathways, like altered pathways in MDM2i-treated cells. Notably, the number of significant genes contributing to both the p53-associated pathways and the BETi-related pathways increased markedly under combination therapy (Figure. 3E and 3F). The induced synergistic response has been attributed to both the indirect and direct activation of P53 from BETi and MDM2i respectively (17).

Signature analysis of the gene in DMSO, BETi, MDM2i and Combination treatment.

In the multi-differential expression analysis, a total of 1,295 significant genes were identified across all group comparisons. There was an overlap of 822 significant genes between the combination-treated and BETi-treated samples (Figure. 4A), and these genes were either upregulated or downregulated (Supplementary Figure. 1I). A fold-fold plot also demonstrated a positive correlation of 0.7 between gene expression changes in the combination and BETi treatments, indicating that these treatments induce similar transcriptional responses (Figure. 4B). The BET inhibitor alone showed a moderate positive correlation with the combination treatment, suggesting partially shared effects along with some treatment-specific differences. Finally, the combination treatment exhibited less similarity to the MDM2i treatment, indicating that the major combination's transcriptional effects are driven by BET inhibition rather than MDM2 inhibition (Figure. 4B).

Overall, the significant genes showed three distinct expression signatures across the treatment groups (Figure. 4C). The first signature consisted of genes that were lowly expressed in all treatments, with the combination group showing the strongest reduction among the three. The second signature included genes that were downregulated in both the BETi and combination treatments, while remaining unchanged in the MDM2i group (Figure. 4D). The last, signature comprised genes that were highly expressed in the BETi-treated and combination-treated samples (Figure. 4D & E). This signature indicated a coordinated activation of p53-mediated stress and apoptotic responses, alongside metabolic stress pathways such as hypoxia, starvation, and oxidative stress (Figure. 4F). The genes like *BBC3 (PUMA)* in the first enriched pathway of the signature 3, are mediators of AML cells apoptosis.(Figure. 4G)

In summary, this analysis demonstrates that BET inhibitors induce cell-cycle arrest by suppressing the transcription of genes essential for AML cell proliferation and division, such as *CTSG* and *LYZ*, indirectly activating p53-related pathways. In contrast, MDM2 inhibition activates p53, leading to upregulation of genes such as *CDKN1A* and promoting apoptosis. The combination treatment influences genes and pathways involved in cell-cycle arrest, apoptosis, and proliferation more profoundly than either treatment alone (Figure. 4H). This aligns with the findings of Latif et al., who showed that

the synergistic killing effect of combined MDM2i and BETi therapy arises from increased activation of p53 target genes, such as *PUMA* and *NOXA*, resulting from the dual action of p53 stabilization by MDM2 inhibition and the release of BRD4-mediated repression through BET inhibition (17).

This analysis is limited by the lack of discussion of two of the three identified signatures, as well as the absence of a Cellular Component Gene Ontology analysis, which could have provided additional insight into the cellular structures and processes underlying the observed biological effects.

References

1. Jiang H, Wang Y, Wang C, Yang L, Wang S, Wang F, et al. Cathepsin D promotes acute myeloid leukemia progression through stabilization of the anti-apoptotic proteins. *Cell Death Dis.* 2025 Aug 12;16(1):611. doi:10.1038/s41419-025-07949-7
2. Handschuh L, Kaźmierczak M, Milewski M, Giźalski M, Łuczak M, Wojtaszewska M, et al. Gene expression profiling of acute myeloid leukemia samples from adult patients with AML-M1 and -M2 through boutique microarrays, real-time PCR and droplet digital PCR. *Int J Oncol.* 2017 Dec 28. doi:10.3892/ijo.2017.4233
3. Rostami M, Kharajo RS, Parsa-kondelaji M, Ayatollahi H, Sheikhi M, Keramati MR. Altered expression of NEAT1 variants and P53, PTEN, and BCL-2 genes in patients with acute myeloid leukemia. *Leuk Res.* 2022 Apr 1;115:106807. doi:10.1016/j.leukres.2022.106807 PubMed PMID: 35231756.
4. Yan H, Wang Z, Sun Y, Hu L, Bu P. Cytoplasmic NEAT1 Suppresses AML Stem Cell Self-Renewal and Leukemogenesis through Inactivation of Wnt Signaling. *Advanced Science.* 2021 Nov 5;8(22). doi:10.1002/advs.202100914
5. Alaseem AM. Advancements in MDM2 inhibition: Clinical and pre-clinical investigations of combination therapeutic regimens. *Saudi Pharmaceutical Journal.* 2023 Oct;31(10):101790. doi:10.1016/j.jsps.2023.101790
6. Lopes MR, Machado-Neto JA, Traina F, Campos P de M, Saad STO, Favaro P. Differential profile of CDKN1A and TP53 expressions in bone marrow mesenchymal stromal cells from myeloid neoplasms. *Rev Bras Hematol Hemoter.* 2016 Oct;38(4):368–70. doi:10.1016/j.bjhh.2016.07.002
7. Dutto I, Tillhon M, Cazzalini O, Stivala LA, Prosperi E. Biology of the cell cycle inhibitor p21CDKN1A: molecular mechanisms and relevance in chemical toxicology. *Arch Toxicol.* 2015 Feb 17;89(2):155–78. doi:10.1007/s00204-014-1430-4
8. Adams PD, Latif AL, Newcombe A, Robertson N, Terradas M, Stewart H, et al. BET Inhibitors Potentiate Activation of p53 and Killing of AML By MDM2 Inhibitors — a Candidate Combination Therapy. *Blood.* 2018 Nov 29;132(Supplement 1):3912–3912. doi:10.1182/blood-2018-99-115509
9. Djamai H, Berrou J, Dupont M, Coudé MM, Delord M, Clappier E, et al. Biological Effects of BET Inhibition by OTX015 (MK-8628) and JQ1 in NPM1-Mutated (NPM1c)

- Acute Myeloid Leukemia (AML). *Biomedicines*. 2021 Nov 17;9(11):1704. doi:10.3390/biomedicines9111704
10. Lindström MS, Bartek J, Maya-Mendoza A. p53 at the crossroad of DNA replication and ribosome biogenesis stress pathways. *Cell Death Differ*. 2022 May 20;29(5):972–82. doi:10.1038/s41418-022-00999-w
 11. Latif AL, Newcombe A, Li S, Gilroy K, Robertson NA, Lei X, et al. BRD4-mediated repression of p53 is a target for combination therapy in AML. *Nat Commun*. 2021 Jan 11;12(1):241. doi:10.1038/s41467-020-20378-8
 12. Zhu Y, Dai H, Wang Y, Liang Y, Feng W, Yuan Y. Targeting FEN1 Suppresses the Proliferation of Chronic Myeloid Leukemia Cells Through Regulating Alternative End-Joining Pathways. *DNA Cell Biol*. 2021 Aug 1;40(8):1101–11. doi:10.1089/dna.2021.0239
 13. Zhang R, Liu Z, Zhang G. CDC45 modulates MCM7 expression and inhibits cell proliferation by suppressing the PI3K/AKT pathway in acute myeloid leukemia. *Am J Transl Res*. 2021 Sep 15;13:10218–32.
 14. Zhang S, Zhao Y, Heaster TM, Fischer MA, Stengel KR, Zhou X, et al. BET inhibitors reduce cell size and induce reversible cell cycle arrest in AML. *J Cell Biochem*. 2019 May 11;120(5):7309–22. doi:10.1002/jcb.28005
 15. Kamel WA, Krishnaraj J, Ohki R. The Role of PHLDA3 in Cancer Progression and Its Potential as a Therapeutic Target. *Cancers (Basel)*. 2025 Mar 22;17(7):1069. doi:10.3390/cancers17071069
 16. Luo LF, Hou CC, Yang WX. Nuclear factors: Roles related to mitochondrial deafness. *Gene*. 2013 May 15;520(2):79–89. doi:10.1016/j.gene.2013.03.041 PubMed PMID: 23510774.
 17. Li S, Latif AL, Newcombe A, Gilrory K, Robertson N, Finlay D, et al. 3102 – A SYNTHETIC LETHALITY APPROACH TO ERADICATE AML VIA SYNERGISTIC ACTIVATION OF PRO-APOPTOTIC P53 BY MDM2 AND BET INHIBITORS. *Exp Hematol*. 2020 Aug;88:S70. doi:10.1016/j.exphem.2020.09.115